Who looks after what? - complete text version

The three main cloud service models are Infrastructure as a Service, or IaaS, Platform as a Service, or PaaS, and Software as a Service, or SaaS. Each one is a different balance of control, convenience and responsibility. In this activity you will read six short descriptions of cloud tools and services. For each one, you will work out who looks after what, the provider or the customer’s own team, and then choose the service model it fits. You will finish by matching each model to the people who use it. For each description, first decide who looks after each of three layers. The pattern you find points to the model. Then choose the model. If a choice is not right, the feedback explains why and you can try again. At the end, match the three models to their typical users. The businesses are fictional. This activity is not scored, and nothing you choose or write is stored or sent anywhere unless you save a text file yourself. Allow about 20 minutes.

How to use this version

You can do every stage on paper or in a document. Read each description, decide for each of the three layers whether the provider or the customer’s own team looks after it, then choose the service model. Read the feedback for every option and compare it with your choices. If you chose differently, work out why before moving on. At the end, match the three models to their typical users.

The three service models

The three service models, and what each one means. Keep these in mind as you read.

- Infrastructure as a Service, or IaaS: The provider supplies virtual servers and storage. The customer’s team manages the operating systems, software and security settings.
- Platform as a Service, or PaaS: The provider supplies a managed platform, with the servers, tools and environment already set up. Developers write, test and run their own applications on it.
- Software as a Service, or SaaS: The provider supplies a finished application over the internet and operates the underlying service. Customers use the application while still managing their own data, identities, access and configuration within it.

To tell the models apart, ask who operates each of three service layers.

- Physical servers and storage
- Operating systems, runtimes and development tools
- The application and its updates

Description 1 of 6: Virtual data-centre hosting

A mid-sized engineering firm wants to host its design files and internal applications without maintaining physical servers. Its technology team still needs to choose operating systems, install software and manage security settings. The firm only wants to rent the virtual machines and storage space needed to support its existing workflows.

A. Who looks after what?

Who looks after: Physical servers and storage? Options: The provider; The customer’s own team.

- Physical servers and storage, The provider - Best fit. Feedback: That fits. The firm wants to host its files and applications without maintaining physical servers, so the provider looks after the hardware.
- Physical servers and storage, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The firm wants to host its files and applications without maintaining physical servers, so the provider looks after the hardware.

Who looks after: Operating systems, runtimes and development tools? Options: The provider; The customer’s own team.

- Operating systems, runtimes and development tools, The provider - Not the best fit. Feedback: Not quite. It is the customer’s own team. The firm’s technology team still chooses operating systems and manages security settings.
- Operating systems, runtimes and development tools, The customer’s own team - Best fit. Feedback: That fits. The firm’s technology team still chooses operating systems and manages security settings.

Who looks after: The application and its updates? Options: The provider; The customer’s own team.

- The application and its updates, The provider - Not the best fit. Feedback: Not quite. It is the customer’s own team. The firm runs its own internal applications and installs its own software.
- The application and its updates, The customer’s own team - Best fit. Feedback: That fits. The firm runs its own internal applications and installs its own software.

Look at what you decided. In the infrastructure model, the provider operates the infrastructure layer. In the platform model, it operates the infrastructure and platform layers. In the software model, it operates the infrastructure, platform and application service layers. In every model, customers still keep responsibility for their own data, identities, access and configuration.

B. Which service model fits?

Now choose the service model this description fits.

- IaaS - Best fit. Feedback: That fits. The cloud provides virtual servers and storage, so the firm manages its own systems and security settings while the provider handles the hardware.
- PaaS - Not the best fit. Feedback: Not quite. In PaaS, the customer deploys and maintains its own application on the provider's managed OS and runtime. Here the firm mainly needs computing resources to host and control its existing systems.
- SaaS - Not the best fit. Feedback: Not quite. SaaS delivers complete applications to end users, not the server space and infrastructure this firm needs.

Description 2 of 6: Disaster-recovery environment

A financial institution needs a backup system that can mirror its own data centre in the event of a failure. The system must replicate network configurations and storage layers so operations can resume quickly. The company’s technology department will handle the operating systems and application recovery, but it does not want to maintain physical infrastructure.

A. Who looks after what?

Who looks after: Physical servers and storage? Options: The provider; The customer’s own team.

- Physical servers and storage, The provider - Best fit. Feedback: That fits. The institution does not want to maintain physical infrastructure, so the provider looks after the hardware.
- Physical servers and storage, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The institution does not want to maintain physical infrastructure, so the provider looks after the hardware.

Who looks after: Operating systems, runtimes and development tools? Options: The provider; The customer’s own team.

- Operating systems, runtimes and development tools, The provider - Not the best fit. Feedback: Not quite. It is the customer’s own team. The institution’s technology department will handle the operating systems.
- Operating systems, runtimes and development tools, The customer’s own team - Best fit. Feedback: That fits. The institution’s technology department will handle the operating systems.

Who looks after: The application and its updates? Options: The provider; The customer’s own team.

- The application and its updates, The provider - Not the best fit. Feedback: Not quite. It is the customer’s own team. The institution’s technology department will handle application recovery.
- The application and its updates, The customer’s own team - Best fit. Feedback: That fits. The institution’s technology department will handle application recovery.

Look at what you decided. In the infrastructure model, the provider operates the infrastructure layer. In the platform model, it operates the infrastructure and platform layers. In the software model, it operates the infrastructure, platform and application service layers. In every model, customers still keep responsibility for their own data, identities, access and configuration.

B. Which service model fits?

Now choose the service model this description fits.

- IaaS - Best fit. Feedback: That fits. The cloud provides virtual data centres and backup storage that mirror the institution’s systems. The technology team still manages recovery and configuration, but not the physical hardware, which makes this a clear IaaS solution.
- PaaS - Not the best fit. Feedback: Not quite. PaaS would host a customer-deployed application on a managed OS and runtime. This scenario is about reproducing servers, storage and networking for disaster recovery, which is infrastructure control.
- SaaS - Not the best fit. Feedback: Not quite. SaaS means using the provider's application. Here the organisation needs its own recovery environment with control over infrastructure resources, not just access to a provider application.

Description 3 of 6: Application-development workspace

A startup will deploy its own logistics application to a cloud environment where the provider operates the servers, operating system and runtime. The team manages the application code and data, but does not want to maintain the underlying platform.

A. Who looks after what?

Who looks after: Physical servers and storage? Options: The provider; The customer’s own team.

- Physical servers and storage, The provider - Best fit. Feedback: That fits. The developers do not want to set up or maintain servers, so the provider looks after the hardware.
- Physical servers and storage, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The developers do not want to set up or maintain servers, so the provider looks after the hardware.

Who looks after: Operating systems, runtimes and development tools? Options: The provider; The customer’s own team.

- Operating systems, runtimes and development tools, The provider - Best fit. Feedback: That fits. The provider operates the servers, operating system and runtime that the startup deploys its application onto.
- Operating systems, runtimes and development tools, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The provider operates the servers, operating system and runtime that the startup deploys its application onto.

Who looks after: The application and its updates? Options: The provider; The customer’s own team.

- The application and its updates, The provider - Not the best fit. Feedback: Not quite. It is the customer’s own team. The developers write and test the logistics app themselves.
- The application and its updates, The customer’s own team - Best fit. Feedback: That fits. The developers write and test the logistics app themselves.

Look at what you decided. In the infrastructure model, the provider operates the infrastructure layer. In the platform model, it operates the infrastructure and platform layers. In the software model, it operates the infrastructure, platform and application service layers. In every model, customers still keep responsibility for their own data, identities, access and configuration.

B. Which service model fits?

Now choose the service model this description fits.

- IaaS - Not the best fit. Feedback: Not quite. In IaaS, the customer manages the operating system, runtime and deployed application. Here the provider runs the platform layer, while the startup manages only its application and data.
- PaaS - Best fit. Feedback: That fits. This is PaaS because the startup deploys and maintains its own application, while the provider operates the servers, operating system and runtime underneath.
- SaaS - Not the best fit. Feedback: Not quite. SaaS would mean using the provider's application. Here the startup is deploying and managing its own logistics application on the provider's managed platform.

Description 4 of 6: Business-analytics toolkit

An analytics team deploys its own analytics application to an online environment. The provider operates the servers, operating system and runtime, and scales capacity automatically. The team maintains the analytics application, connections and dashboards without managing the underlying platform.

A. Who looks after what?

Who looks after: Physical servers and storage? Options: The provider; The customer’s own team.

- Physical servers and storage, The provider - Best fit. Feedback: That fits. The provider manages the infrastructure.
- Physical servers and storage, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The provider manages the infrastructure.

Who looks after: Operating systems, runtimes and development tools? Options: The provider; The customer’s own team.

- Operating systems, runtimes and development tools, The provider - Best fit. Feedback: That fits. The provider operates the servers, operating system and runtime, and scales the platform for the deployed analytics application.
- Operating systems, runtimes and development tools, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The provider operates the servers, operating system and runtime, and scales the platform for the deployed analytics application.

Who looks after: The application and its updates? Options: The provider; The customer’s own team.

- The application and its updates, The provider - Not the best fit. Feedback: Not quite. It is the customer’s own team. The team maintains its analytics application, including its connections, analyses and dashboards, while the provider runs the underlying platform.
- The application and its updates, The customer’s own team - Best fit. Feedback: That fits. The team maintains its analytics application, including its connections, analyses and dashboards, while the provider runs the underlying platform.

Look at what you decided. In the infrastructure model, the provider operates the infrastructure layer. In the platform model, it operates the infrastructure and platform layers. In the software model, it operates the infrastructure, platform and application service layers. In every model, customers still keep responsibility for their own data, identities, access and configuration.

B. Which service model fits?

Now choose the service model this description fits.

- IaaS - Not the best fit. Feedback: Not quite. In IaaS, the team would manage the operating system, runtime and analytics application itself. Here the provider operates that platform layer, while the team maintains only its analytics application.

- PaaS - Best fit. Feedback: That fits. This is PaaS because the team deploys and maintains its analytics application, while the provider operates the servers, operating system and runtime and handles scaling.
- SaaS - Not the best fit. Feedback: Not quite. SaaS would mean using the provider's analytics application. Here the team maintains its own analytics application on a provider-operated platform.

Description 5 of 6: Customer-relationship management tool

A regional sales company adopts an online customer-management system where employees log in through a browser to manage leads, track communications and generate reports. The vendor operates and maintains the application service and its underlying cloud components, while the company manages its own customer data, user access and configuration.

A. Who looks after what?

Who looks after: Physical servers and storage? Options: The provider; The customer’s own team.

- Physical servers and storage, The provider - Best fit. Feedback: That fits. The system is fully managed by the vendor, so the vendor looks after the hardware.
- Physical servers and storage, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The system is fully managed by the vendor, so the vendor looks after the hardware.

Who looks after: Operating systems, runtimes and development tools? Options: The provider; The customer’s own team.

- Operating systems, runtimes and development tools, The provider - Best fit. Feedback: That fits. The vendor operates the platform components that support the customer-management application.
- Operating systems, runtimes and development tools, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The vendor operates the platform components that support the customer-management application.

Who looks after: The application and its updates? Options: The provider; The customer’s own team.

- The application and its updates, The provider - Best fit. Feedback: That fits. The vendor handles updates, and employees simply log in through a browser.
- The application and its updates, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The vendor handles updates, and employees simply log in through a browser.

Look at what you decided. In the infrastructure model, the provider operates the infrastructure layer. In the platform model, it operates the infrastructure and platform layers. In the software model, it operates the infrastructure, platform and application service layers. In every model, customers still keep responsibility for their own data, identities, access and configuration.

B. Which service model fits?

Now choose the service model this description fits.

- IaaS - Not the best fit. Feedback: Not quite. IaaS would require the sales team to install and maintain their own software, which they are not doing here.
- PaaS - Not the best fit. Feedback: Not quite. This scenario describes the company using the vendor's finished customer-management application. With PaaS, the customer would deploy its own application onto a provider-managed runtime instead.
- SaaS - Best fit. Feedback: That fits. The company is using a finished customer-management application operated by the vendor. Staff access the tool through a browser, while the company still manages its own data, user access and configuration within it.

Description 6 of 6: HR and payroll suite

A human-resources department uses a subscription-based HR application to manage employee records, payroll and performance reviews. Staff access the system online, while the provider maintains the application service and rolls out updates; the organisation still manages its own employee data, user access and configuration.

A. Who looks after what?

Who looks after: Physical servers and storage? Options: The provider; The customer’s own team.

- Physical servers and storage, The provider - Best fit. Feedback: That fits. The provider operates the underlying computing infrastructure that the HR application runs on.
- Physical servers and storage, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The provider operates the underlying computing infrastructure that the HR application runs on.

Who looks after: Operating systems, runtimes and development tools? Options: The provider; The customer’s own team.

- Operating systems, runtimes and development tools, The provider - Best fit. Feedback: That fits. The provider operates the platform components that support the HR application service.

- Operating systems, runtimes and development tools, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The provider operates the platform components that support the HR application service.

Who looks after: The application and its updates? Options: The provider; The customer’s own team.

- The application and its updates, The provider - Best fit. Feedback: That fits. The provider rolls out new features automatically.
- The application and its updates, The customer’s own team - Not the best fit. Feedback: Not quite. It is the provider. The provider rolls out new features automatically.

Look at what you decided. In the infrastructure model, the provider operates the infrastructure layer. In the platform model, it operates the infrastructure and platform layers. In the software model, it operates the infrastructure, platform and application service layers. In every model, customers still keep responsibility for their own data, identities, access and configuration.

B. Which service model fits?

Now choose the service model this description fits.

- IaaS - Not the best fit. Feedback: Not quite. This scenario is about using a finished HR application. With IaaS, the organisation would rent computing resources and still manage its own operating systems and applications on them.
- PaaS - Not the best fit. Feedback: Not quite. PaaS would mean the organisation deploys its own HR application onto a provider-managed runtime. Here, the provider is maintaining the HR application itself for staff to use.
- SaaS - Best fit. Feedback: That fits. The cloud delivers a ready-to-use HR and payroll application that staff use online. The provider maintains the service, while the organisation still manages its own employee data, user access and configuration within the application.

Match the model to its users

Each model suits a different kind of user. Choose the service model that fits each group.

Group 1 of 3

Technology teams that need to manage their own systems and security, but not the hardware.

- IaaS - Best fit. Feedback: That fits. Infrastructure as a Service: The provider supplies virtual servers and storage. The customer’s team manages the operating systems, software and security settings.

- PaaS - Not the best fit. Feedback: Not quite. Platform as a Service: The provider supplies a managed platform, with the servers, tools and environment already set up. Developers write, test and run their own applications on it. This group is teams that manage systems themselves.
- SaaS - Not the best fit. Feedback: Not quite. Software as a Service: The provider supplies a finished application over the internet and operates the underlying service. Customers use the application while still managing their own data, identities, access and configuration within it. This group is teams that manage systems themselves.

Group 2 of 3

Teams who want to deploy and run their own application on a provider-managed runtime without managing the underlying servers and platform software.

- IaaS - Not the best fit. Feedback: Not quite. Infrastructure as a Service: The provider supplies virtual servers and storage. The customer's team manages the operating systems, software and security settings. This group is building applications on a managed platform.
- PaaS - Best fit. Feedback: That fits. Platform as a Service: The provider supplies a managed platform, with the servers, tools and environment already set up. Developers write, test and run their own applications on it.
- SaaS - Not the best fit. Feedback: Not quite. Software as a Service: The provider supplies a finished application over the internet and operates the underlying service. Customers use the application while still managing their own data, identities, access and configuration within it. This group is building applications on a managed platform.

Group 3 of 3

People who want to use a finished application provided by the vendor, while their organisation still manages its own data, access and configuration in that tool.

- IaaS - Not the best fit. Feedback: Not quite. Infrastructure as a Service: The provider supplies virtual servers and storage. The customer's team manages the operating systems, software and security settings. This group is using a finished application.
- PaaS - Not the best fit. Feedback: Not quite. Platform as a Service: The provider supplies a managed platform, with the servers, tools and environment already set up. Developers write, test and run their own applications on it. This group is using a finished application.
- SaaS - Best fit. Feedback: That fits. Software as a Service: The provider supplies a finished application over the internet and operates the underlying service. Customers use the application while still managing their own data, identities, access and configuration within it.

Reflect and save

You sorted six descriptions into service models. Here is a summary of the model you chose for each.

- Virtual data-centre hosting: Infrastructure as a Service, or IaaS.
- Disaster-recovery environment: Infrastructure as a Service, or IaaS.
- Application-development workspace: Platform as a Service, or PaaS.
- Business-analytics toolkit: Platform as a Service, or PaaS.
- Customer-relationship management tool: Software as a Service, or SaaS.
- HR and payroll suite: Software as a Service, or SaaS.

Optional reflection: Choose one pair of models that is easy to confuse, such as platform and software. What question could you ask a customer to find out which one they need? Reflection and saving a copy are optional. You have completed the activity after sorting all six descriptions and matching all three user groups. In the interactive version, Save my summary downloads the six models and typical user groups; it includes your reflection if you added one. The service models differ in how much the customer manages and who uses the result. Knowing that lets you match each model to the level of control and convenience a customer needs.